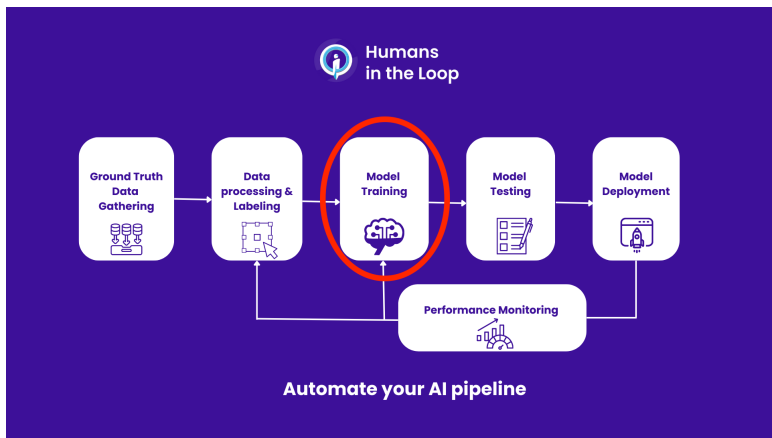


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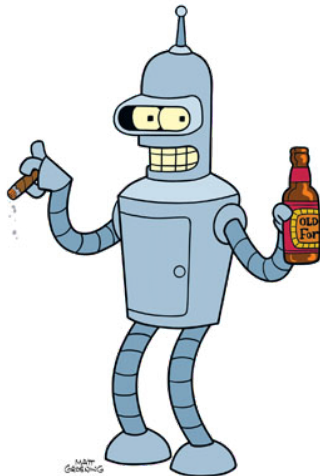
Christopher Zorn

September 28, 2026

“The Loop”



Programming vs. Learning



Some terms:

- Outcome / Target: The “thing” the ML is attempting to learn/replicate. This may be a *continuous* variable or a *discrete* one, or may be undefined.
- Features / Predictors / Covariates / Inputs: The aspects of the data that are used to inform the machine’s decision about the target.
- Model: The set of algorithms and parameters that are recursively adapted (“fitted”) to the data.
- Training Data: The data that are used to “train” the model.
- Test Data: Data that are used to validate / “test” the model after it learns.

Flavors of Machine Learning

1. **Supervised**

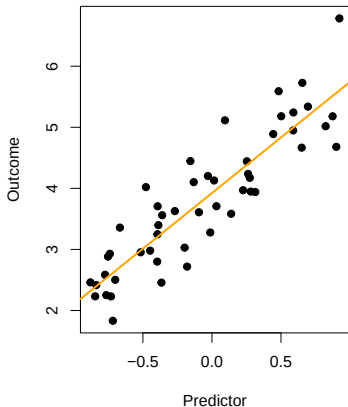
(1.5) **(Semi-Supervised)**

2. **Unsupervised**

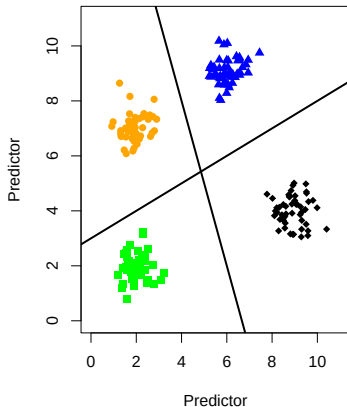
3. **Reinforcement**

Supervised ML: Continuous and Discrete

Regression



Classification

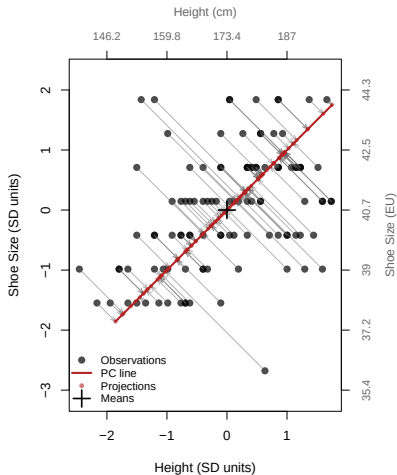
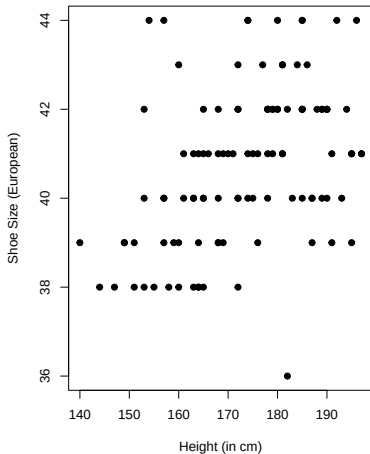


Supervised Learning

Traits:

- Begins with / requires labeled data (defined target)
- Maps predictors to labels via flexible functions (“parameters”)
- High accuracy, high interpretability, easy to validate / evaluate
- Requires large amounts of (labeled) data; prone to overfitting
- Examples: Regression (linear, nonlinear, ridge/lasso/etc.), Logistic Regression, Decision Trees, Random forests, kNN, SVM

Unsupervised Learning



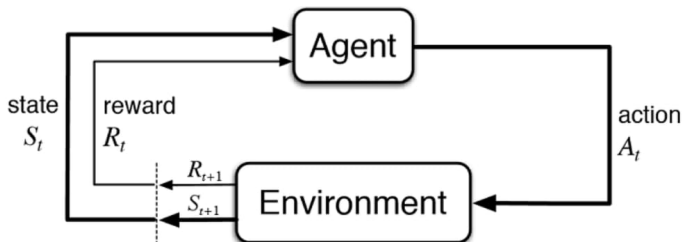
(source)

Unsupervised Learning

Traits:

- Works with unlabeled data (no measured “target” variable / *latent* outcome)
- Model finds + extracts “hidden” patterns / structures in the data
- Related to “dimension reduction” in multivariate statistics
- Works better than supervised learning on unstructured data
- Generally harder to assess, less accurate, less interpretable than supervised learning
- Examples: Principal components analysis (PCA), cluster analysis (*k*-means, etc.), LDA, others...

Reinforcement Learning



(source)

Reinforcement Learning is:

- Cyclical + adaptive
- Goal-Based
- Good at complex tasks, long-term performance
- Less accurate in the short term

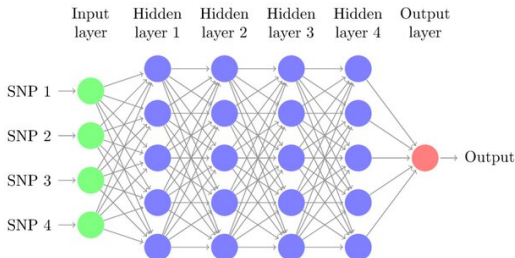
Variant I: Ensembles

Details:

- Idea: Average predictions from multiple models
- Advantages: Better accuracy, lower overfitting, very versatile
- Disadvantages: Complexity = higher computational cost, lower interpretability



Variant II: “Deep” Learning



It's about *architecture*:

- “Shallow” models: a single “layer” (one model between input → output)
- “Deep” models: First layer feeds into additional layers...
- Useful for learning from unstructured / unlabeled data

Variant III: Transfer Learning

It's a *strategy*.

- Concept: take knowledge gained from one task or domain and apply it to a different but related task or domain
- Usually: pretrain a model on a large / general dataset, then fine-tune it on a smaller targeted dataset
- Useful when (a) problems are similar, and (b) (targeted) data are scarce
- Usable with **all types of ML** (+ combinations)

Variant IV: “One-Shot” Learning

Idea:



- Build a model that can “recognize” something based on a single data point
- Often uses transfer learning + similarity metrics (e.g., images)
- Use cases: facial recognition, character / handwriting recognition, etc.

Key Things About ML (from Domingos 2012)

Thing:

1. The goal is *generalization*.
2. Data isn't enough: every model has assumptions.
3. Overfitting is (very) bad.
4. More data is good.
5. Choosing predictors ("features") is key.
6. Ensembles > single models.